



MANIPAL

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – 5_Theory

Reg. Number	Name	Branch
261100690006	MOHAMMED HARIS	Cybersecurity

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MANIPAL SCHOOL OF INFORMATION SCIENCES

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Write a Bash script that accepts a filename as the first command-line argument.

Check whether an argument is provided.

Check whether the file exists using a file operator.

If the file exists, display the first 5 lines using head.

Otherwise, display an appropriate error message.

File Type and Permission Checker

```

haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ nano sample.txt
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ ./ex1.sh sample.txt
First 5 lines of the file:
something
next line
line 3
line 4
ls
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ cat ex1.sh
#!/bin/bash
if [ -z "$1" ]
then
echo "Error: No filename provided."
exit 1
fi
if [ -e "$1" ]
then
echo "First 5 lines of the file:"
head -n 5 "$1"
else
echo "Error: File does not exist."
fi
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ 

```

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

a regular file,

a directory,

readable,

writable, and

executable.

Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ cat check.sh
#!/bin/bash

if [ -z "$1" ]; then
    echo "Error: No path provided."
    exit 1
fi

path="$1"

if [ -f "$path" ]; then
    echo "$path is a regular file."
elif [ -d "$path" ]; then
    echo "$path is a directory."
else
    echo "$path is neither a regular file nor a directory."
fi

if [ -r "$path" ]; then
    echo "$path is readable."
else
    echo "$path is not readable."
fi

if [ -w "$path" ]; then
    echo "$path is writable."
else
    echo "$path is not writable."
fi

if [ -x "$path" ]; then
    echo "$path is executable."
else
    echo "$path is not executable."
fi

haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash check.sh sample.txt
sample.txt is a regular file.
sample.txt is readable.
sample.txt is writable.
sample.txt is not executable.
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash check.sh ~
/home/haris is a directory.
/home/haris is readable.
/home/haris is writable.
/home/haris is executable.
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$
```

Write a Bash script that accepts a log file as a command-line argument and searches for the words:

failed, denied, unauthorized, and error.

```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ cat search.sh
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi
if [ ! -f "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi
echo "Matching records:"
grep -iE "failed|denied|unauthorized|error" "$1"
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash search.sh sample.log
Matching records:
2026-09-18 08:17:45 ERROR Database connection failed.
2026-09-18 08:18:12 WARN Access denied for user bob.
2026-09-18 08:20:05 ERROR Unable to read configuration file.
2026-09-18 08:21:44 WARN Unauthorized access attempt detected.
2026-09-18 08:23:56 ERROR Authentication failed for user charlie.
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$
```

**Write a Bash script that accepts a filename and checks whether:
the file exists,
it is a regular file,
it is empty or non-empty.
If it is non-empty, display the first 5 lines.**

```
haris@ubuntu-arn:~/Documents/LoS/Theory/Th5$ cat filecheck.sh
#!/bin/bash
if [ -z "$1" ]
then
echo "Error: No filename provided."
exit 1
fi
if [ ! -e "$1" ]
then
echo "Error: File does not exist."
exit 1
fi
echo "File exists."
if [ -f "$1" ]
then
echo "It is a regular file."
else
echo "It is not a regular file."
exit 1
fi
if [ -s "$1" ]
then
echo "File is non-empty."
echo "First 5 lines:"
head -n 5 "$1"
else
echo "File is empty."
fi
haris@ubuntu-arn:~/Documents/LoS/Theory/Th5$ bash filecheck.sh sample.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
something
next line
line 3
line 4
15
haris@ubuntu-arn:~/Documents/LoS/Theory/Th5$ bash filecheck.sh empty.txt
File exists.
It is a regular file.
File is empty.
haris@ubuntu-arn:~/Documents/LoS/Theory/Th5$
```

**Write a Bash script named investigate.sh that accepts a log file as a command-line argument.
The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.
Display the first 5 matching records and report the total number of matching records.**

```
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ cat investigate.sh
#!/bin/bash
if [ -z "$1" ]
then
echo "Error: No log file provided."
exit 1
fi
if [ ! -e "$1" ]
then
echo "Error: Log file does not exist."
exit 1
fi
if [ ! -r "$1" ]
then
echo "Error: Log file is not readable."
exit 1
fi
read -p "Enter search term: " term
echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5
count=$(grep -ic "$term" "$1")
echo "Total matching records: $count"
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash investigate.sh sample.log
Enter search term: denied
First 5 matching records:
2026-09-18 08:18:12 WARN Access denied for user bob.
Total matching records: 1
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash investigate.sh sample.log
Enter search term: error
First 5 matching records:
2026-09-18 08:17:45 ERROR Database connection failed.
2026-09-18 08:20:05 ERROR Unable to read configuration file.
2026-09-18 08:23:56 ERROR Authentication failed for user charlie.
Total matching records: 3
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash investigate.sh samp.log
Error: Log file does not exist.
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$ bash investigate.sh
Error: No log file provided.
haris@ubuntu-arm:~/Documents/LoS/Theory/Th5$
```